2D Systolic Array with Hybrid Pruning, Approximate Multiplier and Region-Aware Activation for CNN Acceleration

ECE 284 Fall25 Project

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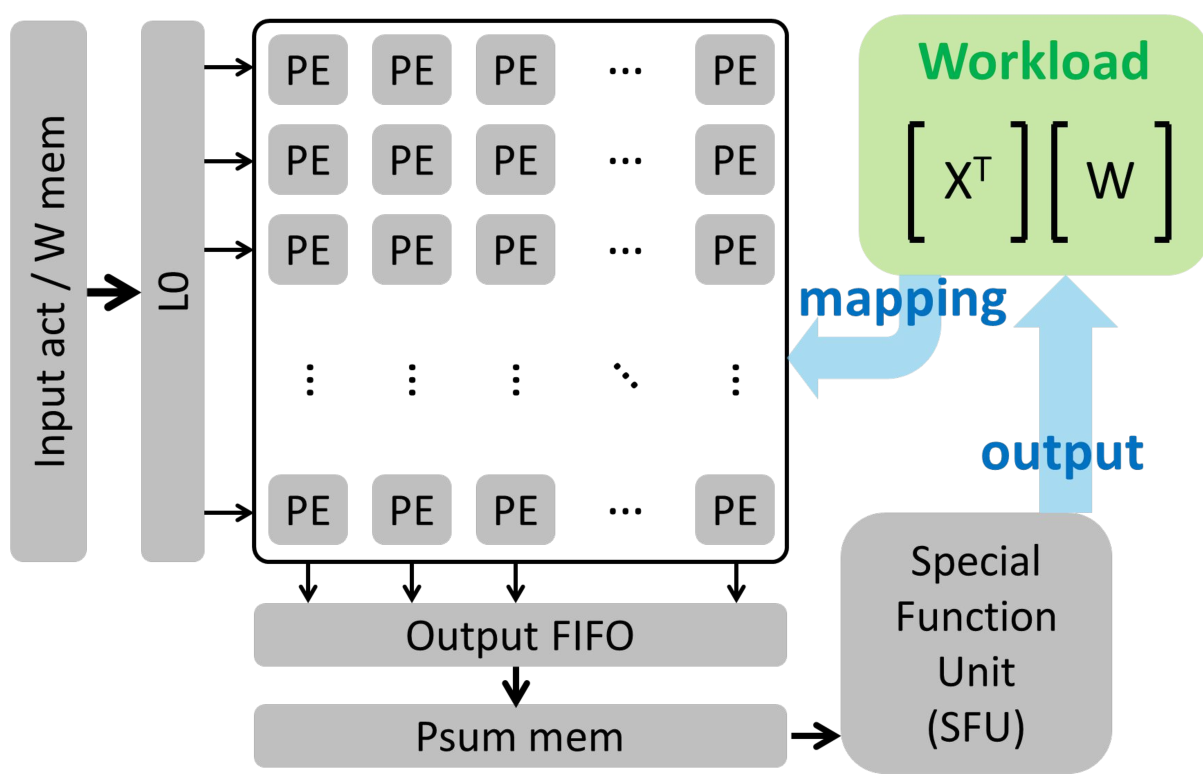
Motivation & Basic 2D Systolic Array

Neural networks, which form the foundation of AI, are mainly consisted of **General Matrix Multiplication (GEMM)** operations. To accelerate GEMM, **2D systolic arrays** have been widely adopted due to their high data reuse, high frequency enabled by their pipelined structure, and efficient data movement.

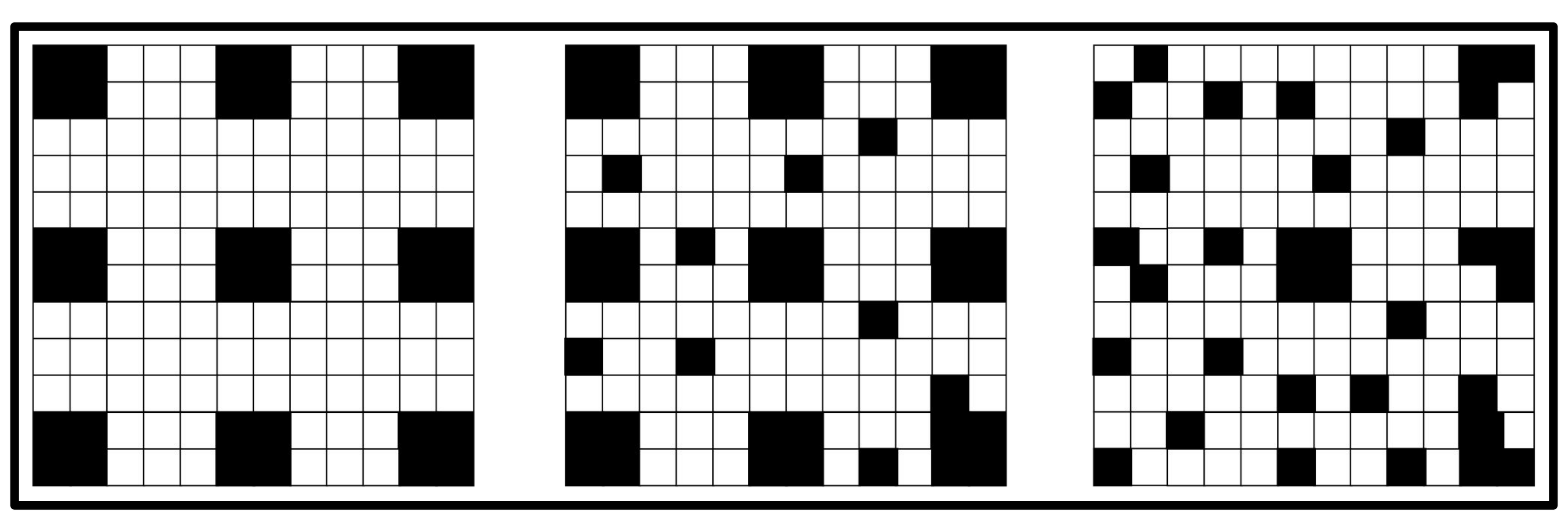
In this work, we propose a method to utilize such 2D systolic arrays **even more effectively**.



Alpha 1: Hybrid Pruning (Structure + Unstructured)

Hybrid Pruning keeps advantages from both Structure and Unstructured Pruning

- **Structured Efficiency**
  - Easier to design efficient hardware with structured pruning
- **Retain same levels of accuracy**
  - Reduces model size and compute while maintaining accuracy



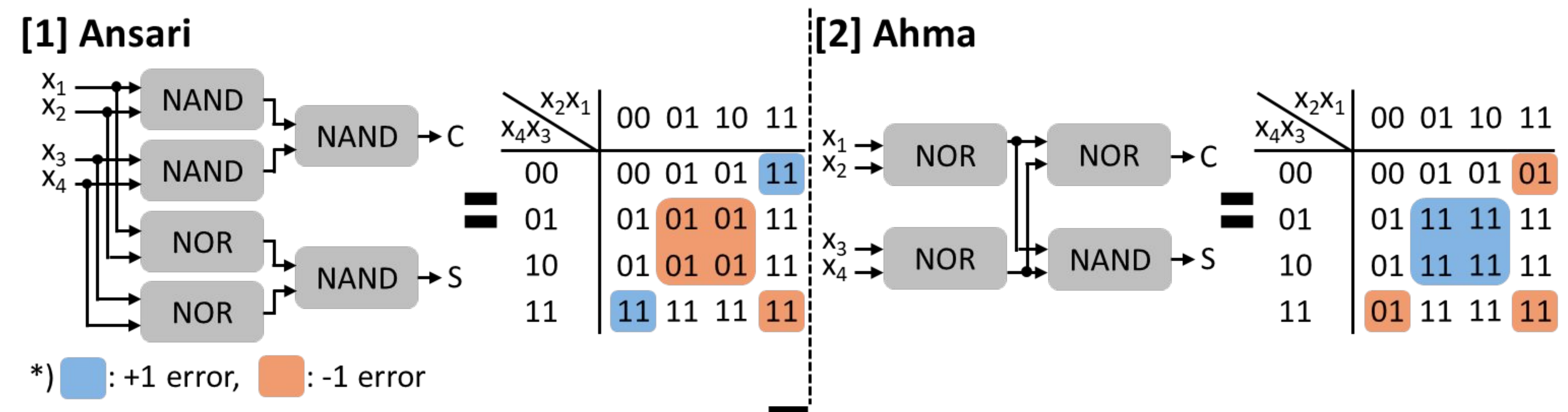
- Testing with VGG16 on CIFAR-10 dataset
  - Original accuracy is 93%, we prune all layers to 80%.
  - Prune and retrain until no significant improvement

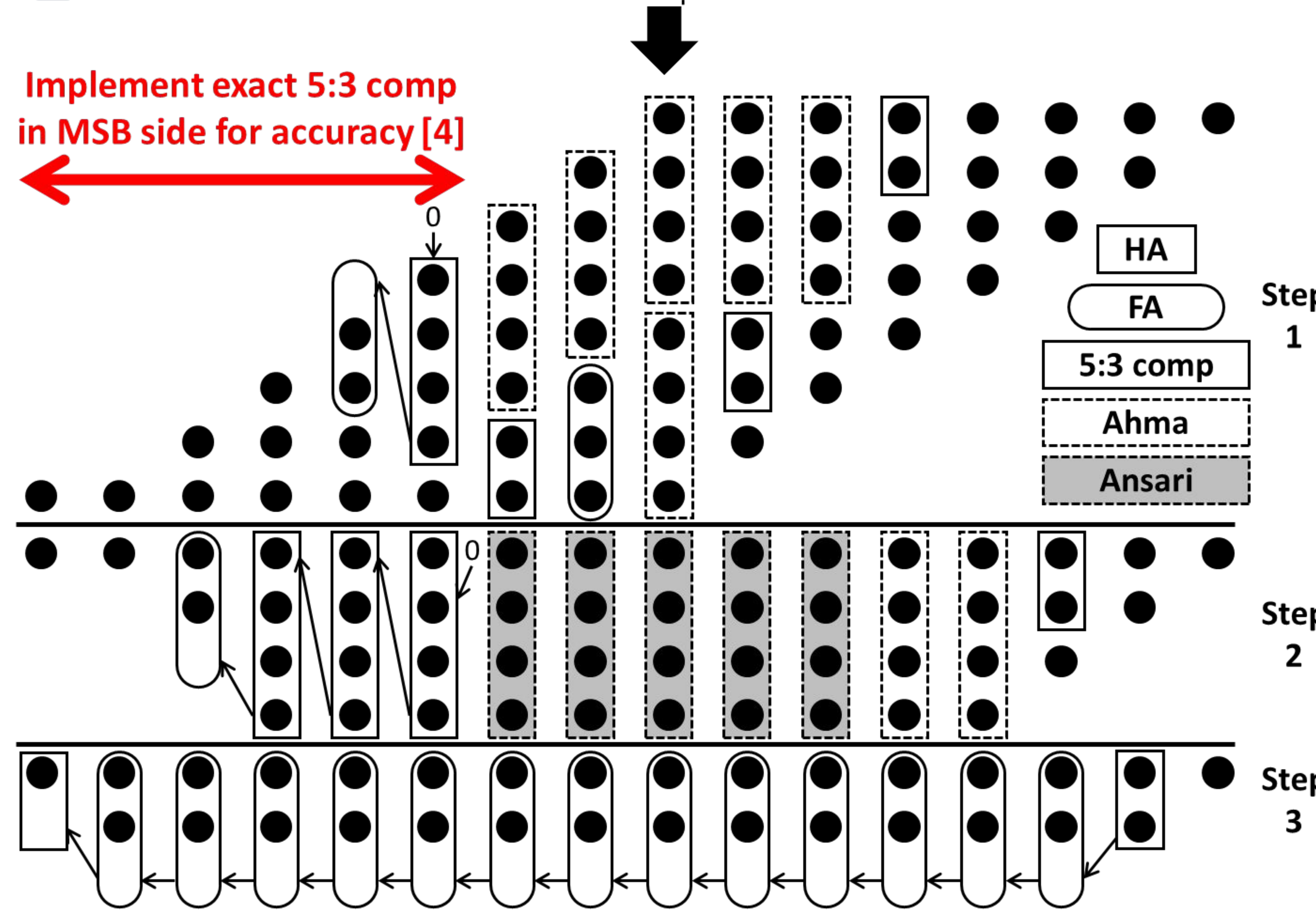
	Structured	Hybrid	Unstructured
After Pruning	10%	10%	10%
Retrain	71.01%	81.52%	83.37%

Alpha 2: Approximate Multiplier for Low Power

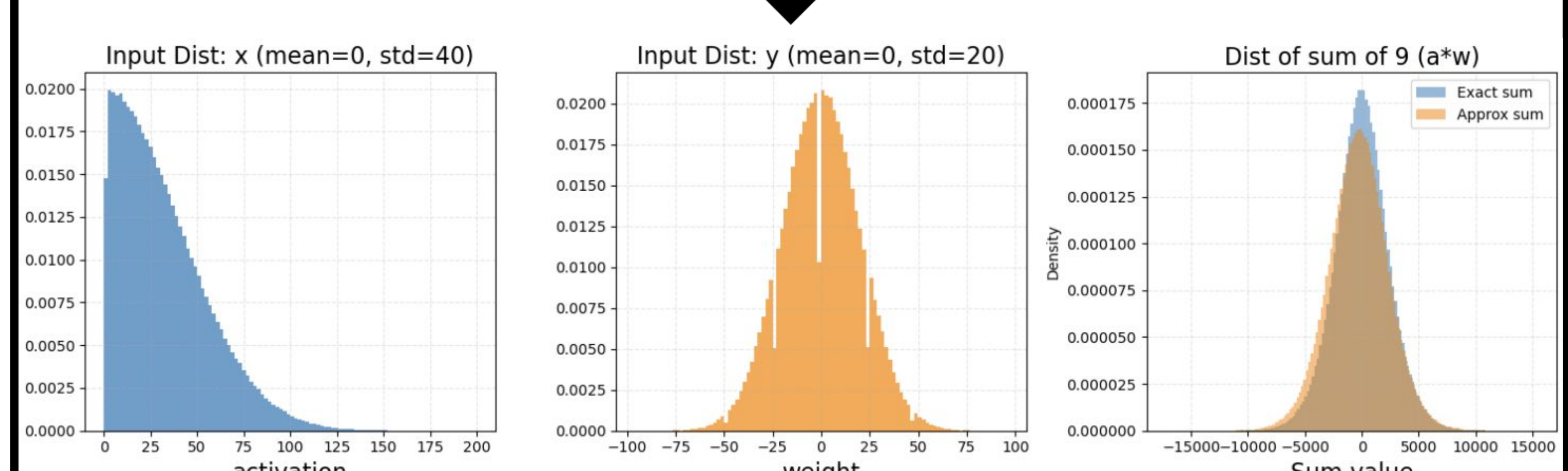
- We use two 4:2 compressors composed of NOR and NAND, whose error distributions are symmetric w.r.t. each other. [3]

(4:2 compressor: adds four 1-bit inputs and makes a 2-bit output)





<8×8 signed Dadda multiplier>



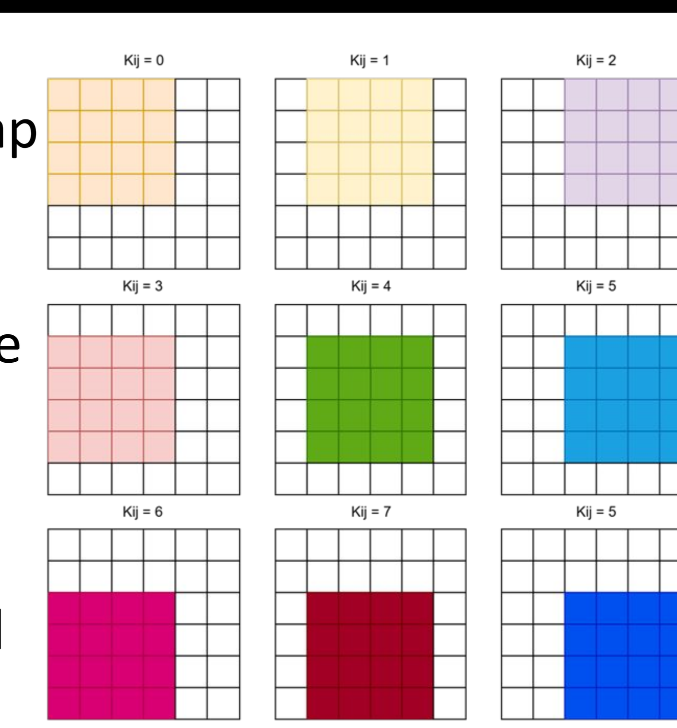
Alpha 3: Region-Aware Activation Extraction

For each convolution position only a specific region of the input feature map is actually used.

Our design identifies the exact valid window for each and provides only the required activations.

Our method reduces calculation from **36 * 9 = 324 to 16 * 9 = 144.**

This alpha saves **44%** of operation and **26%** of total time



Alpha 4: Hybrid Read Mode and Pipelined RD/WR

L0 Read Signal Mode

Case 1: Propagate

Transfering **activation.**

Case 2: Simultaneous

Transferring **weight.**

Pipeline FIFO Design

Allowing FIFO loads and write data at the same time

FIFO depth lowers from **64 to 16.**

-> Alpha 3 saves 32% of cycles in total

Alpha 5: Zero Skipping MAC Operation

When an activation or weight is zero, we gated FF clock and MAC.

1. **Gating the a/b/c flip-flops:**
   - No power savings.
2. **Gating the MAC inputs:**
   - Achieved **23.7 % power reduction**.

```
assign a_pad = {1'b0, a};
assign product = zero? 0 : a_pad * b;

assign psum = zero? c : product + c;
assign out = psum;
```

Final Result with FPGA Mapping

Cyclone IV GX						
	Fmax	Dynamic Power	Total Elements	TOPS	TOPS/Watt	TOPS/Area
Vanilla	132.21 MHz	25.69 mW	12,897	16922 ops/s	52824 TOPS/W	1312 TOPS/elements
Alpha	109.7 MHz	21.79 mW	13,831	14041 ops/s	44638 TOPS/W	1015 TOPS/elements

Reference

[1] M. S. Ansari, et al, "Low-Power Approximate Multipliers Using Encoded Partial Products and Approximate Compressors," JETCAS, vol. 8, no. 3, pp. 404-416, Sept. 2018

[2] M. Ahmadinejad, M. H. Moaiyeri, and F. Sabetzadeh, "Energy and area efficient imprecise compressors for approximate multiplication at nanoscale," AEU-Int. J. Electron. Commun., vol. 110, Oct. 2019

[3] HyunJin Kim, et al, "Low-biased probabilistic multipliers using complementary inaccurate compressors", DSP, Vol 146, 2024

[4] A. G. M. Strollo, et al, "Comparison and Extension of Approximate 4-2 Compressors for Low-Power Approximate Multipliers", TCAS1, vol. 67, no. 9, pp. 3021-3034, Sept. 2020